

A Joint Evaluation of Deployment Efficiency and Semantic Accuracy in Vision–Language Models on Edge Systems



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Motivation

Vision-language models enable multimodal perception for scientific edge applications such as environmental monitoring, urban sensing, and field instrumentation. Most VLM evaluation, however, emphasizes accuracy on standardized benchmarks, leaving a gap in how these models behave under the memory, power, and thermal budgets that define real edge deployments. We jointly evaluate compact VLMs along both axes, deployment efficiency and semantic correctness, across two heterogeneous edge platforms to surface the trade-offs that matter for matching models to specific scientific use cases under tight resource budgets.

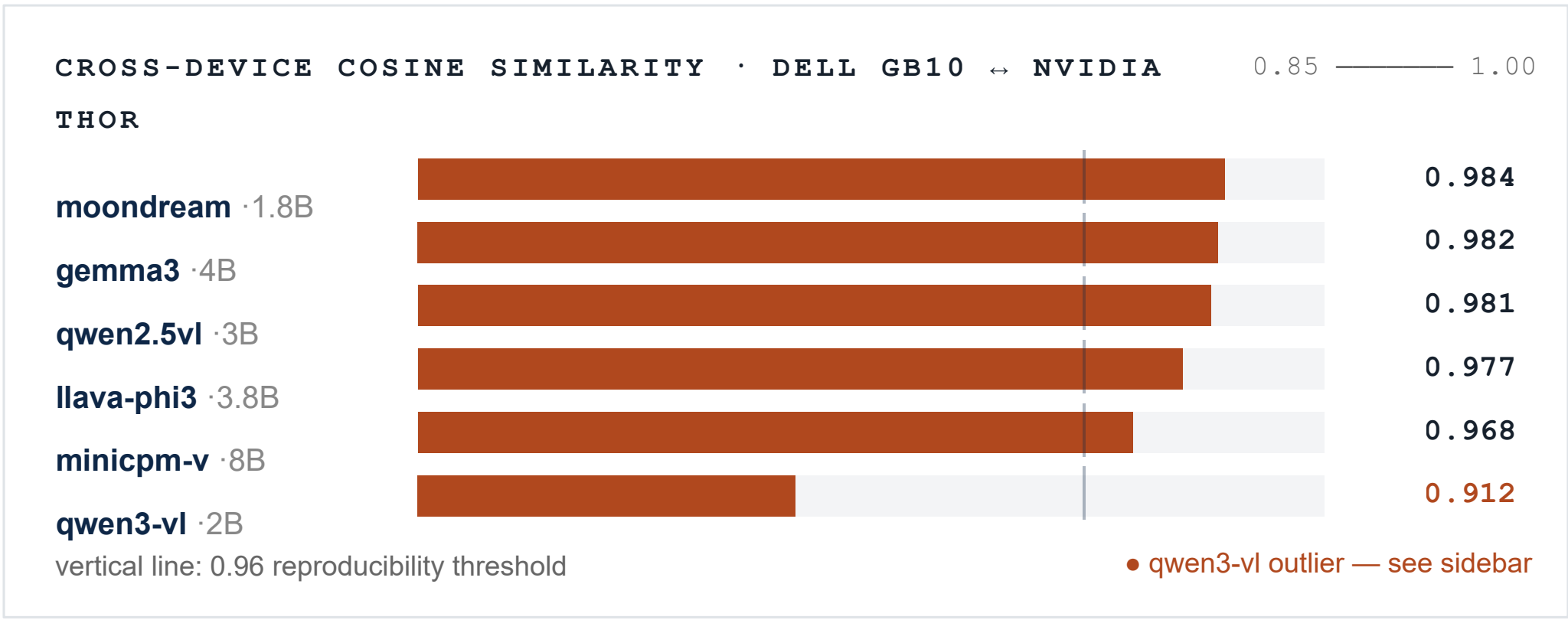


Setup

Models	moondream · llava-phi3 · gemma3 · minicpm-v · qwen2.5vl:3b
Quant.	Q4_K_M (identical weights, both devices)
Hardware	Dell Pro Max GB10 — Grace Blackwell · 128 GB LPDDR5X Jetson AGX Thor (T5000) — Blackwell · 128 GB LPDDR5X
Prompts	single-prompt “Describe...” vs 5-task structured
Scale	100 COCO imgs × 5 tasks × 5 runs × 2 platforms = 30k

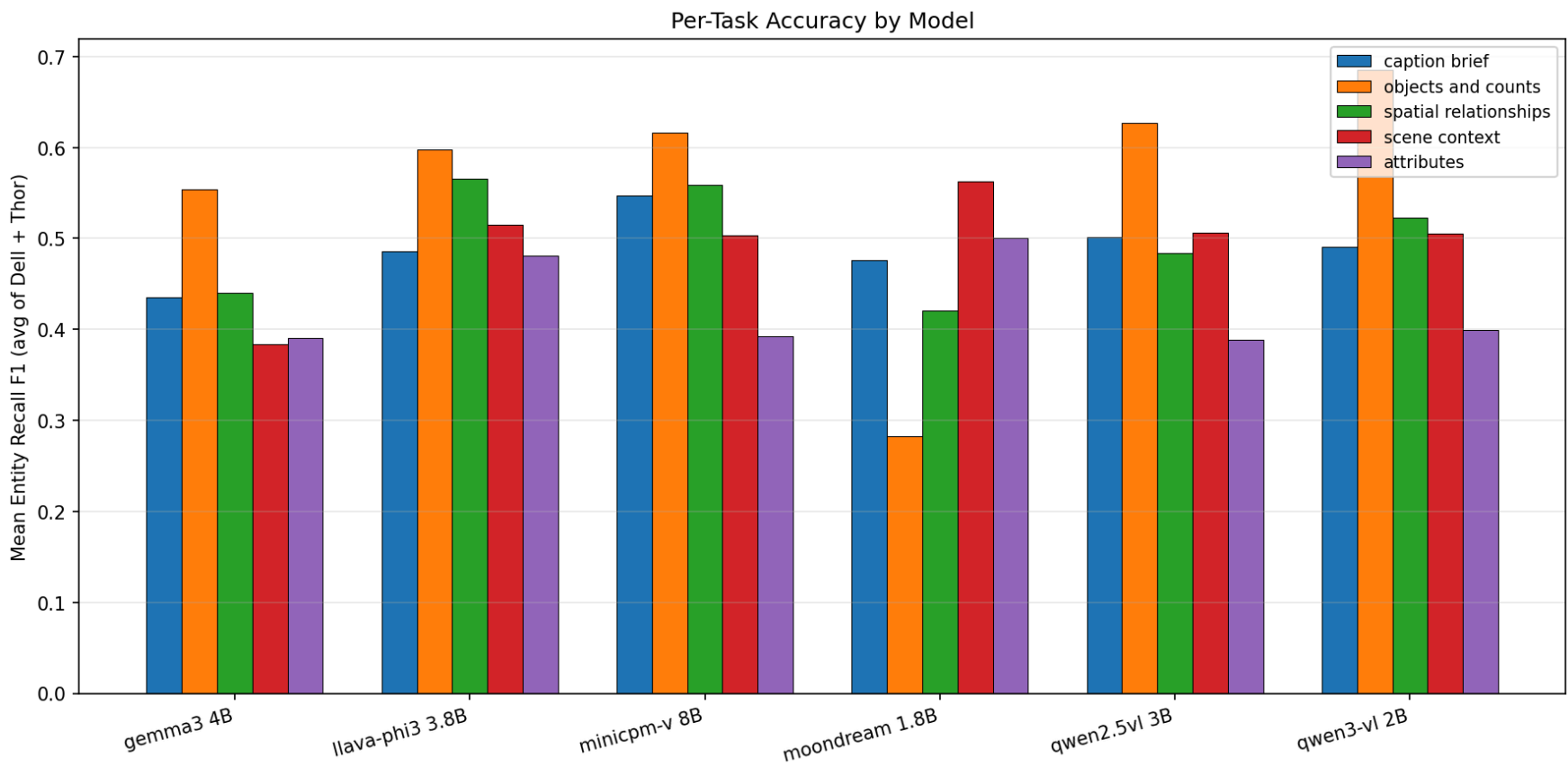
Results

Finding 1. Five of six models cluster between 0.96 and 0.98 cross-device cosine similarity. Within-device stability is 0.99+ on both platforms. Reproducibility holds across every task in the 5-task workload.



Variance also reproduces. Each model's tail ratio (p99/p50) is nearly identical on Dell and Thor, moondream 2.21/2.19, minicpm-v 1.98/1.93, gemma3 1.14/1.23. The same models exhibit consistent behavior across both devices, both in their outputs and in their runtime performance.

Accuracy. minicpm-v ▶ qwen2.5vl ▶ moondream ▶ llava-phi3 ▶ gemma3 ▶ qwen3-vl
Throughput. moondream ▶ qwen3-vl ▶ qwen2.5vl ▶ llava-phi3 ▶ minicpm-v ▶ gemma3
Tail (p99/p50). qwen3-vl ▶ gemma3 ▶ llava-phi3 ▶ qwen2.5vl ▶ minicpm-v ▶ moondream



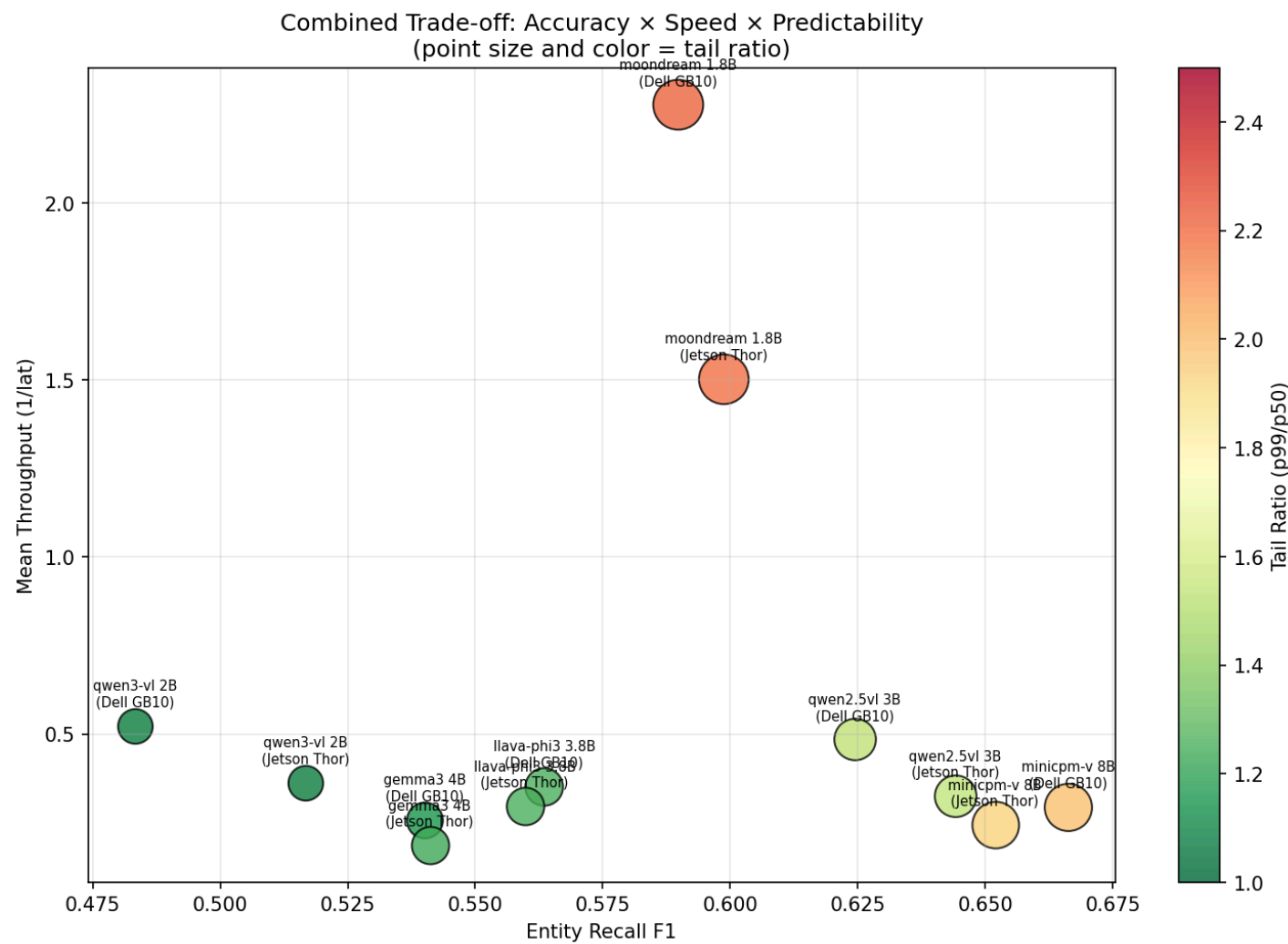
Per-task accuracy, 5-task workload (both devices averaged). minicpm-v dominates overall; qwen3-vl dominates object counting at 0.685; moondream collapses on enumeration.

Model	Device	n images	n runs	Latency (ms)	Throughput (img/s)	Power (W)	Tokens /sec	TPS /W	J /Token	Total Tokens
gemma3 4B	Dell GB10	500	2,500	1314.5	0.7607	41.45	71.7	1.73	0.772	180,242
gemma3 4B	Jetson Thor	500	2,500	1916.9	0.5217	27.27	59.5	2.18	0.743	180,521
llava-phi3 3.8B	Dell GB10	500	2,500	785.3	1.2734	44.91	75.8	1.69	0.655	138,730
llava-phi3 3.8B	Jetson Thor	500	2,500	938.2	1.0659	36.06	67.7	1.88	0.640	139,200
minicpm-v 8B	Dell GB10	500	2,500	1495.4	0.6687	46.06	45.3	0.98	1.156	150,865
minicpm-v 8B	Jetson Thor	500	2,500	1796.0	0.5568	35.69	43.7	1.22	1.089	150,431
moondream 1.8B	Dell GB10	497	2,485	225.8	4.4286	31.87	163.5	5.13	0.303	66,014
moondream 1.8B	Jetson Thor	500	2,500	349.2	2.8638	23.49	126.4	5.38	0.411	66,911
qwen2.5-vl 3B	Dell GB10	500	2,500	585.6	1.7075	35.04	95.7	2.73	0.534	100,599
qwen2.5-vl 3B	Jetson Thor	500	2,500	946.5	1.0565	25.93	70.7	2.72	0.644	101,244
qwen3-vl 2B	Dell GB10	500	2,500	646.6	1.5466	38.69	151.4	3.91	0.354	186,271
qwen3-vl 2B	Jetson Thor	500	2,500	1037.3	0.9641	25.17	122.5	4.87	0.347	202,845

Methods

- Generation:** deterministic (temperature 0, fixed seed); llama.cpp Q4_K_M weights.
- Cross-device similarity:** sentence-transformer embeddings of paired outputs, cosine similarity.
- Latency tail:** measured per-image, per-run; reported as p99/p50.
- Empty outputs:** generations with zero non-whitespace tokens.
- Rationale:** settings chosen to minimize generation variability and isolate platform behavior in the Dell–Thor comparison.

Finding 2. Three orthogonal axes, three different winners. The 5-task data adds a fourth axis, task-dependent strengths.



Accuracy / speed / tail trade-off across both workloads. moondream is fastest by an order of magnitude; minicpm-v is most accurate; qwen3-vl has the tightest tail.

Dell Pro Max (GB10)				Jetson AGX Thor			
GB10 Grace Blackwell · 128 GB LPDDR5X · DGX OS 7				Blackwell GPU · 128 GB LPDDR5X · JetPack 7			
Benchmark averages over 6 models (moondream · qwen2.5-vl · qwen3-vl · llava-phi3 · gemma3 · minicpm-v), 5-task regime, 100 COCO images each.							
METRIC	DELL PRO MAX (GB10)			JETSON AGX THOR			Δ
Latency seconds per inference	0.84 s			1.16 s			+38%
Throughput inferences per second	1.73			1.17			−32%
Peak system memory GB	20.0 GB			53.1 GB			—
Power draw W	39.7 W			28.9 W			−27%
Energy / inference	33.4 J			33.7 J			+1%
Tokens / sec decode rate	100.6			81.7			−19%
J / Token energy efficiency	0.629			0.646			+3%

Values summarize benchmark averages for the two platforms. Power telemetry is platform-specific (Dell: NVML GPU draw; Thor: tegrastats VDD_GPU rail), so power and energy are not directly comparable. Peak system memory reflects system/unified memory usage.

Key Takeaways & Future Work

Cross-device reproducibility holds (0.96–0.98 for 5 of 6 models; tail latency transfers too), but no single model dominates accuracy, speed, and predictability rank differently, and tasks favor different models. Single-prompt benchmarks miss real failures (qwen3-vl: 69% empty outputs). Energy per inference stays ~constant (~33 J) despite Dell leading on speed and Thor on power. Future work: probe qwen3-vl's prompt sensitivity and test multi-model execution under SAGE-realistic contention

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